Do Macro and Micro RLVR Interventions Compose? A Pre-Registered Factorial Study of Reasoning Breadth and Solution Diversity in a Small Language Model

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***Abstract*—Reinforcement learning with verifiable rewards (RLVR) raises the single-sample accuracy of language models, but whether it shrinks the set of problems a model can solve is disputed, the evidence is automatic, and fixes that act within a rollout group and across prompts have not been tested together. We study all three on Qwen2.5-0.5B-Instruct trained with GRPO on MATH. A pre-registered 2×3 factorial crosses a macro factor (none, or a topic budget on positive advantage) with a micro factor (none, the SELF filter, or redistribution over solution modes), with four or five seeds per cell (27 models), plus a blinded human audit. At 2.0 rollouts per training problem, vanilla GRPO does not shrink the boundary: MATH-500 Pass@1 moves from 30.3% to 30.4% and Pass@32 from 73.0% to 74.4%, and boundary exits stay at the sampling-noise level. Neither pre-registered contrast is significant (95% CIs of [-1.7, $+2.1$] and [-1.2, $+2.9$] points on the gain from $k = 1$ to $k = 32$). The topic budget raises Pass@1 by about one point over vanilla (uncorrected). Annotators agreed on 85% of solution pairs (PABAK $+0.70$) and judged 50 of 51 agreed pairs to share one approach; the word-overlap partitioner that splits them into "modes" scores 55%, below the 98% of always answering "same". The diversity these fixes reweight is mostly wording. A tie-breaking third annotator found lost problems no more often validly solved than kept ones. We hypothesise that the null reflects low training intensity; we did not vary it.**

Code, data and pre-registration: **https://github.com/2005legend/Reasoning-boundary-paradox-of-SLM**

***Index Terms*—reinforcement learning with verifiable rewards, GRPO, Pass@k, small language models, human evaluation**

I. Introduction

Group Relative Policy Optimization (GRPO) [1] and related RLVR methods are now the standard way to improve mathematical reasoning in open language models. Whether they teach a model to solve new problems, or only to solve solvable ones more reliably, is disputed. Yue et al. [2] found that base models overtake their RLVR-trained versions when many samples are drawn, a *Pass@k inversion* later analysed as a shrinking reasoning boundary [3]–[6]. Others report the opposite. Prolonged training expands the boundary [7], RLVR comes out ahead when only correct chains of thought are counted [8], and a two-stage account reconciles the two by training duration [9]. A third view blames the measurement: aggregate declines track how few rollouts each training problem receives [10], and Pass@k at large k rewards lucky hits [11].

The evidence on every side is automatic. Pass@k cannot tell a valid solution from a lucky guess. CoT-Pass@k hands that judgment to LLM judges, which accept corrupted reasoning almost as often as clean reasoning [12], and approach-level diversity has been scored by an LLM judge calibrated on 80 human-labelled pairs [13]. As far as we know, no study has asked people what RLVR loses: whether the solutions that disappear are distinct strategies, and whether lost problems were ever solved by valid reasoning.

A third gap concerns the fixes. *Intra-group* methods reshape credit using the diversity or structure of one prompt's rollouts, for example by giving rare correct modes more credit than common ones [14]–[19], and *cross-prompt* methods reweight whole problems or tasks [3], [10], [20], [21]. The two act on different axes but have not been tested together (ReCo combines two corrections, both within one prompt [15]). One pre-registered study addresses all three gaps. Our contributions:

- *Is shrinkage real at small-language-model (SLM) scale?* A multi-seed measurement on Qwen2.5-0.5B-Instruct [22] trained on MATH [23], with problem-level boundary entries and exits.
- *What is lost, according to people?* A blinded audit of (H1) whether our partitioner's solution modes are distinct strategies and (H2) whether the base model's successes

on lost problems were valid reasoning.

- *Do the fixes compose?* A 2×3 factorial crossing a topic-level budget on positive advantage (CB, macro) with no micro term, the SELF filter, or intra-group mode-rarity redistribution (MEG).
- The answers: at about two rollouts per training problem RLVR expands the boundary slightly, no fix or combination moves the shrinkage slope by more than about one point per unit $\ln k$, and the modes the fixes rely on are mostly wording. In exploratory comparisons (uncorrected), the topic budget raises Pass@1 by about one point over vanilla, and CB×SELF ranks first on every Pass@k.

II. Related Work

Is shrinkage real? Yue et al. [2] showed that base models overtake RLVR-trained ones at large k. Nguyen et al. [3] attribute this to negative interference and winner-take-all reinforcement and propose SELF, which trains only on problems whose greedy answer is wrong. Wu et al. [5] argue that RLVR stays largely confined to the base model's support, Zhou and Li [6] place the contraction at the first decision of a solution, and Zhou [4] diagnoses the inversion. Against this, ProRL [7] reports expansion after prolonged, KL-controlled training, Wen et al. [8] report expansion under CoT-Pass@k, and Yao et al. [9] describe an exploitation stage that shrinks the boundary followed by an exploration stage that expands it. Yuan et al. [10] tie the decline to few rollouts per problem and gate problems by their estimated contribution to the boundary (BBG), and Dragoi et al. [11] show that Pass@k favours lucky hits. Other work changes the optimizer, the rollout search or the advantage function [24]–[26]. At SLM scale, GRPO dynamics [27], difficulty scaling [28] and reward granularity [29] have been studied, but not shrinkage.

Measuring what is lost. Lee et al. [13] find that embedding similarity tracks phrasing more than strategy and word-bigram overlap tracks approach better, and that 0.7% of GSM8K problems but 33% of MATH training problems admit more than one approach. Yang et al. [30] code the strategies of frontier models with AI coders and human adjudication, without RL. Taşaltı et al. [12] find that CoT-Pass@k judges accept corrupted chains, and that Pass@k exceeds CoT-Pass@k by 19.7 points on Qwen2.5-generation solvers. Our audit puts people directly on RLVR outcomes.

Intra-group credit redistribution. Cue-GRPO [14] redistributes positive credit over a partition of the correct rollouts by rarity and restores the mean; we adopt its rule as our micro term. ReCo [15] reweights against concentration at the response and token level, and GCPO [16], DRA-GRPO [17], EDAS [18] and TreeAdv [19] reshape advantages using diversity or structure within the group.

Cross-prompt reweighting. MT-GRPO [21] reweights explicit tasks for worst-task reliability, CurveRL [20] reweights prompts by pass-rate distribution, BBG gates problems by their estimated contribution to the boundary [10], and SELF selects by greedy failure [3]. CB works at the level of topics, uses each topic's share of positive advantage as its signal, and keeps state across steps. We ask whether a topic-level and an intra-group intervention compose, not which gate is best.

III. Method

A. GRPO with gated advantages

For a prompt x the policy π_θ samples G completions $y_1, \ldots, y_G$ with binary rewards $r_i \in \{0, 1\}$. GRPO [1] uses the group-normalized advantage $\hat{A}_i = (r_i - \mathrm{mean}(r))/(\mathrm{std}(r) + \epsilon)$. Every condition multiplies it by a non-negative gate weight, $\tilde{A}_i = g_i \hat{A}_i$ with $g_i = g_i^{\mathrm{mac}} g_i^{\mathrm{mic}}$, and minimizes the PPO-clip loss [31] averaged over all T completion tokens in the batch [32]:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_i \sum_{t=1}^{|y_i|} \min\big(\rho_{i,t} \tilde{A}_i,\ \mathrm{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon)\, \tilde{A}_i\big), \tag{1}$$

where i runs over all 16×8 completions of a step and $\rho_{i,t}$ is the ratio of the token's probability under the current and the sampling policy. We take one gradient step per batch, so $\rho_{i,t} = 1$, the clip never binds, and the update is the token-level policy gradient; there is no KL penalty, and vanilla GRPO is $g_i \equiv 1$. Fig. 1 shows one training step.

B. Macro gate: topic budget (CB)

Topics are the $C = 7$ MATH subjects. After each step CB updates an exponential moving average of each topic's *spend*, the mean positive advantage of its rollouts in that step: $s_c \leftarrow (1-\eta)\, s_c + \eta\ \mathrm{mean}_{i:c(x_i)=c} \max(\hat{A}_i, 0)$ with $\eta = 0.05$. Negative advantage never uses up a topic's budget. With the overspend ratio $\rho_c = s_c / \frac{1}{C} \sum_{c'} s_{c'}$,

$$\tilde{g}_c = \delta^{\max(0,\ \rho_c - \vartheta)}, \qquad \vartheta = 1.1,\ \delta = 0.18, \tag{2}$$

so a topic at $1.5\times$ the average spend gets half the credit of one within budget. Spend starts at zero and a topic absent from a step keeps its value. The gate acts only on rollouts with positive advantage: their weights are normalised to mean 1 over the step, clipped to $[0.3, 3]$ and normalised again, so CB moves credit between topics instead of lowering the learning rate. Because negative advantages are untouched, a prompt's gated advantages no longer sum to zero, so CB also acts as a prompt-level reweighting. The original specification ($\vartheta = 1.5$, $\delta = 0.98$, down-weighting only) never fired in the pilot, where spend ratios stayed within $[0.38, 1.46]$, so we changed it before any main run and recorded the change in the pre-registration.

C. Micro gates: SELF filter and mode redistribution (MEG)

SELF. Following [3], a prompt whose greedy answer is already correct gets weight $\lambda = 0$ and every other prompt weight 1. The greedy answer is decoded as one extra row of the same generation call. We implement the selection rule only, not SELF's forward-KL term.

MEG. Let $\mathcal{C}$ be the N correct rollouts of a group, partitioned into solution modes, and let $|M(i)|$ be the size of the mode containing rollout i. Following Cue-GRPO [14],

$$w_i = \frac{N\, |M(i)|^{-\alpha}}{\sum_{j \in \mathcal{C}} |M(j)|^{-\alpha}} \quad (i \in \mathcal{C}), \qquad w_i = 1 \quad (i \notin \mathcal{C}), \tag{3}$$

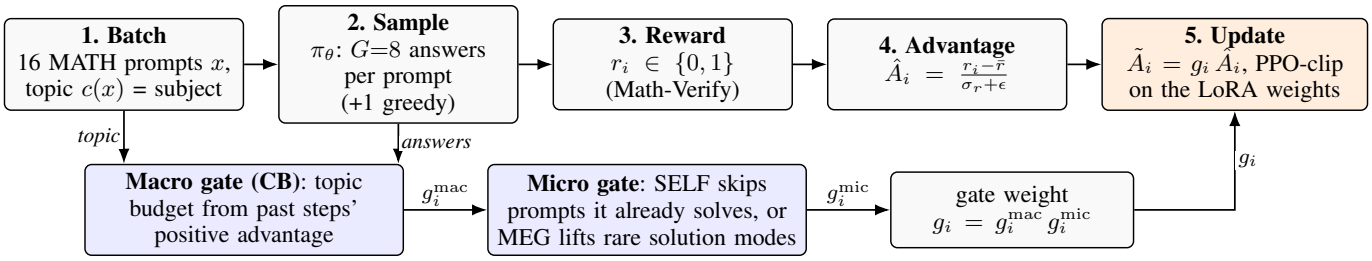


Fig. 1. One training step of gated GRPO. The macro gate knows each prompt's topic and how much positive advantage that topic has absorbed in past steps; the micro gate sees the current group's answers (and greedy answer). Their product scales the advantage. Vanilla GRPO sets both gates to 1.

TABLE I
PARTITIONER CALIBRATION ON BASE-MODEL ROLLOUTS (GROUPS WITH ≥ 4 CORRECT). ⋆ = SELECTED.

MiniLM cosine τ	0.80	0.85	0.90	0.95	
median modes	1.0	1.0	1.0	3.5	
multi-mode groups (%)	2	12	48	88	
Word-bigram threshold	0.30	0.40	0.50	0.60	0.70
median modes	5.0	3.0⋆	2.0	1.0	1.0
multi-mode groups (%)	100	90	60	35	5

TABLE II
THE 2 × 3 FACTORIAL. COMPOSITES MULTIPLY THE MACRO AND MICRO GATES.

macro \ micro	none	SELF filter	MEG redistribution
none	VANILLA	SELF	MEG
CB topic budget	CB	CB×SELF	CB×MEG (HEG-GRPO)

with $\alpha = 0.8$, clipped to $[0.3, 3]$ and rescaled to mean 1 over $\mathcal{C}$ (for $G = 8$ the clip binds only in all-correct groups, whose advantage is zero). In the composites each gate is computed on its own and the weights are multiplied, so under CB×SELF CB's normalisation includes prompts that SELF then drops. MEG changes *which* correct solutions get credit, not how much; an earlier down-weight-only version was confounded with a lower learning rate.

Mode partitioner. Modes come from average-linkage agglomerative clustering of correct completions, on MiniLM (all-MiniLM-L6-v2) [33] sentence embeddings [34] or on word-bigram Jaccard distance. A pre-registered rule, applied to base-model rollouts before training, picks the candidate whose median number of modes per group lies in $[2, 4]$ and is closest to 3. It selected word-bigram Jaccard, threshold 0.4 (median 3.0 modes, 90% of groups with more than one; Table I). The embedding candidates put almost every group's correct rollouts into a single mode, which would make MEG a no-op.

IV. EXPERIMENTAL SETUP

Training. We train Qwen2.5-0.5B-Instruct [22] with LoRA [35] adapters on the MATH training split [23] (7,496 problems with a boxed answer) under the standard "reason step by step and put the final answer in `\boxed{}`" prompt. Table II lists the six conditions and Table III every setting. The reward is binary correctness checked with Math-Verify [36], with a guarded symbolic fallback. The budget is 0.26 epochs, or 2.0 rollouts per training problem. Seed s fixes the data order and LoRA initialization across conditions, so conditions are compared in seed-paired blocks.

TABLE III
TRAINING AND EVALUATION SETTINGS, THE SAME FOR ALL CONDITIONS.

Model	Qwen2.5-0.5B-Instruct, bf16; LoRA r=16, α=32, all projections
Optimizer	AdamW, lr 1×10^{-5}, cosine, 10% warm-up, grad-norm clip 1.0
Batch, steps	16 prompts × G=8 rollouts; 120 steps; MATH levels 1–5
Sampling	T=0.7, top-p 0.95, ≤1,024 new tokens
Loss	PPO-clip ε=0.2, on-policy, token-level mean, no KL
CB / SELF / MEG	η=0.05, ϑ=1.1, δ=0.18 / λ=0 / α=0.8
Evaluation	MATH-500, n=32 samples, T=0.6, top-p 0.95
Hardware	NVIDIA A10G 24 GB (AWS g5.xlarge), one run per GPU

Evaluation and outcome. Every model, including the base model, answers all of MATH-500 [37] 32 times per problem. Pass@k for $k \in \{1, 2, 4, 8, 16, 32\}$ uses the unbiased estimator [38]. No MATH-500 problem appears verbatim in the training split; 4 of the 500 have a near-duplicate there (over 80% shared word 8-grams). The pre-registered primary outcome is the *shrinkage slope*, the least-squares slope of $\Delta(k) = \text{Pass@}k(\text{run}) - \text{Pass@}k(\text{base})$ against $\ln k$ in percentage points per unit $\ln k$; a negative slope means gains at small k turn into losses at large k. We also count boundary *exits* (problems the base model solves at least once in 32 samples and the trained model never does) and *entries* (the reverse).

Pre-registered analysis. Contrasts are linear combinations of condition slopes, paired by seed. The primary contrasts are P1, CB×MEG − CB×SELF (redistribution vs. filtering under the budget), and P2, CB×MEG − MEG (does the macro level add to the micro level?). Secondary contrasts are the two macro×micro interactions (S1, S2), reported as bounds, and the macro main effect (S3); E1 is exploratory. Each contrast has a hierarchical bootstrap interval over seeds and problems (2,000 resamples), and an effect is claimed only if its Holm-adjusted [39] $p < 0.05$ and its interval excludes zero. The pre-registration also fixed how outcomes are worded, including that a vanilla run without shrinkage is reported as such.

Human audit (pre-registered, secondary). Two annotators

TABLE IV
MATH-500 PASS@k (%, MEAN OVER 4 SEEDS FOR VANILLA, SELF, CB AND 5 FOR THE OTHERS), SHRINKAGE SLOPE (PP PER UNIT $\ln k$; HIGHER IS BETTER) AND BOUNDARY ENTRIES/EXITS (OF 500 PROBLEMS). BOLD: BEST MEAN PER COLUMN; ONLY THE PASS@1 GAINS ARE STATISTICALLY SUPPORTED (SEC. V-B).

Condition	P@1	P@4	P@16	P@32	Slope	In / Out
Base model	30.3	50.1	66.3	73.0	–	–
VANILLA	30.4	50.9	67.8	74.4	+0.42	33.2 / 26.0
SELF	30.9	51.1	67.7	74.2	+0.22	32.2 / 26.0
MEG	30.4	50.9	67.2	73.6	+0.15	31.8 / 29.0
CB	31.3	51.4	68.1	74.9	+0.28	36.0 / 26.2
CB×SELF	**31.3**	**51.5**	**68.3**	**75.1**	+0.32	33.6 / **23.0**
CB×MEG	31.0	51.3	68.2	75.0	+0.38	34.0 / 24.0

TABLE V
CONTRASTS ON THE MATH-500 SHRINKAGE SLOPE (PP PER UNIT $\ln k$), PAIRED BY SEED, WITH HIERARCHICAL BOOTSTRAP CIS; THE LAST COLUMN RESTATES THE CI AS POINTS OF GAIN FROM k=1 TO k=32. EACH CONTRAST USES THE SEEDS ITS CONDITIONS SHARE (5 FOR P1, P2; 4 OTHERWISE). p: HOLM-ADJUSTED FOR P1–P2.

	Contrast	Est.	95% CI	p	CI, k=1→32
P1	CB×MEG − CB×SELF	+0.06	[−0.50, +0.61]	0.870	[−1.7, +2.1]
P2	CB×MEG − MEG	+0.23	[−0.34, +0.84]	0.870	[−1.2, +2.9]
S1	CB×MEG interaction	+0.36	[−0.77, +1.51]	0.498	[−2.7, +5.2]
S2	CB×SELF interaction	+0.30	[−0.70, +1.28]	0.568	[−2.4, +4.4]
S3	CB main effect	+0.08	[−0.31, +0.52]	0.670	[−1.1, +1.8]
E1	CB×MEG − Vanilla	−0.06	[−0.72, +0.66]	0.820	[−2.5, +2.3]

who are not authors, Shruti K (BBA) and Sri Prathin (final-year B.E. CSE student), label two blinded sheets without seeing condition, group or partitioner output; as pre-registered, a third non-author annotator labelled only the H2 items on which they disagreed, without seeing their labels. *H1:* 60 pairs of correct base-model solutions to the same training problem, half placed in one mode by the partitioner and half in two, labelled SAME or DIFFERENT approach under the definition of [13]; modes count as approach-level if the partitioner's balanced accuracy against the consensus reaches 70%. *H2:* MATH-500 problems the base model solves but vanilla GRPO never solves in a majority of seeds (*exited*), each matched to a retained problem with the closest base solve count. Annotators judge one correct base-model solution per problem VALID or FLAWED, and the groups are compared with a two-sided Fisher exact test.

Reproducibility. Code, logs, annotation sheets and the script behind every number are in https://github.com/2005legend/Reasoning-boundary-paradox-of-SLM. The pre-registration was written on 2026-10-03 and changed only by dated entries; it was first committed after the runs (`4cdc43a`), so its dates are self-reported. The main study had two or three seeds per cell (16 runs); we then added 11 runs, registered before they ran (14 planned, cut to 11 for budget), giving 27 models and 432,000 scored solutions.

V. RESULTS

A. Shrinkage at 0.5B

Vanilla GRPO does not reproduce the inversion at this budget. Pass@1 moves from 30.3% to 30.4% and Pass@32 rises from 73.0% to 74.4% (Table IV). The curve stays above zero up to $k = 32$, with slope $+0.42$ (Fig. 2, left). On average 33.2 problems enter the boundary and 26.0 leave it, and every other condition also has a positive slope and more entries than exits. Much of this turnover is sampling noise: with 16 samples per problem, the two halves of the base model's own samples already differ by 31 exits and 33 entries, while trained runs show 29–32 exits and 34–38 entries against the base model. Exits are at the noise level and entries exceed it by about five problems.

B. Pre-registered contrasts

All six slopes are positive, from $+0.15$ (MEG) to $+0.42$ (VANILLA). Vanilla's slope is the steepest because its Pass@1 barely moved, so its gains grow with k; the budgeted conditions start higher. The three CB conditions rank first to third on every Pass@k from $k = 1$ to 32, and CB×SELF leads on each and loses the fewest problems (Table IV). Neither primary contrast clears the pre-registered bar (Table V). In points of gain between $k = 1$ and $k = 32$, P1 lies within $[-1.7, +2.1]$ and P2 within $[-1.2, +2.9]$, so any difference between the micro terms is under about three points. The interactions are wider, $[-2.7, +5.2]$ and $[-2.4, +4.4]$, which admits effects larger than vanilla's whole gain at $k = 32$; with four seeds per cell the interaction question stays open. Both primary 90% intervals lie inside ± 1.0 pp per $\ln k$ (± 3.5 points at $k = 32$), so we call P1 and P2 equivalent to zero within that bound, which we chose after seeing the data; the interactions do not meet it. With the between-seed spread we observed, our five seeds detect a difference of about 0.6 (P1) and 0.7 (P2) pp per $\ln k$ at 80% power. We read the primary contrasts as a bounded null, as pre-registered.

Exploratory, not pre-registered. Paired by seed against vanilla (four pairs, t intervals, uncorrected for the 15 comparisons we ran), the topic budget raises Pass@1: by +0.9 points with CB (95% CI [+0.2, +1.5]) and by +1.0 with CB×SELF ([+0.4, +1.6]). No Pass@32 difference is distinguishable from zero, and none of the 15 survives Holm correction (smallest adjusted $p = 0.19$). Since a topic's share of credit does not predict its gain (Sec. V-C), the Pass@1 lift may come from CB's prompt-level reweighting rather than its topic signal; a random-weight control, which we did not run, would separate the two.

C. What each level does

Mode entropy of the correct rollouts starts at 0.60 and does not decay under vanilla GRPO (Fig. 2, right). At evaluation, vanilla's correct answers to a problem fall into 7.44 modes on average, against 6.73 for the base model, and its largest mode holds 39% of them against 43%. There is no collapse for MEG to undo. SELF trains on the 61% of prompts whose greedy answer is wrong and skips the rest, for a mean positive weight

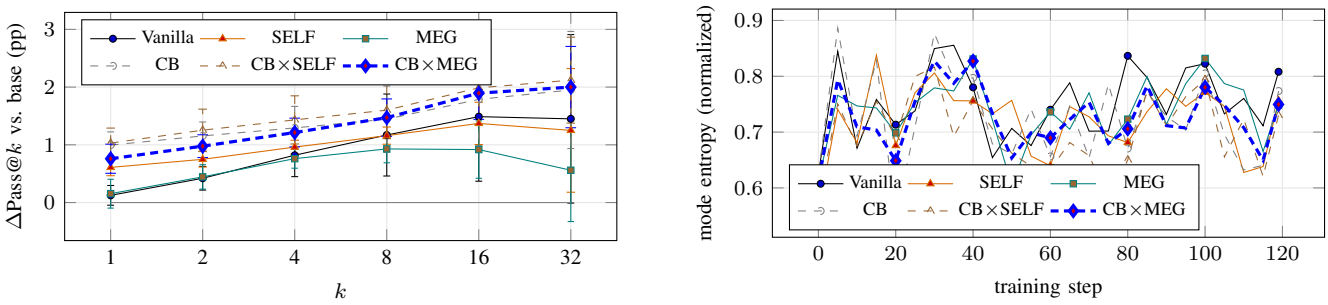


Fig. 2. Left: change in MATH-500 Pass@k relative to the base model (mean ± s.d. over seeds); a curve that fell below zero at large k would mean a shrunk boundary. Right: normalized entropy of the solution-mode distribution among correct rollouts during training (mean over seeds).

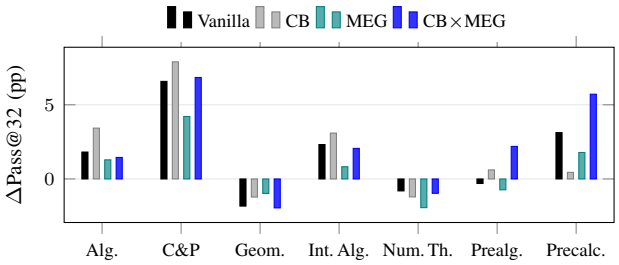


Fig. 3. Change in MATH-500 Pass@32 relative to the base model by subject, the macro gate's topics (mean over seeds).

of 0.31. Changes are uneven across subjects (Fig. 3): under vanilla GRPO, Pass@32 changes by $+6.6$ pp in Counting & Probability and by -1.8 pp in Geometry. The macro gate binds, but not through its topic signal. In the CB runs a subject's share of positive credit does not predict its Pass@32 change (Spearman $\rho = -0.02$, $n = 98$ subject-runs, pooled and so not independent), which leaves the premise of a topic budget unsupported.

Boundary exits barely separate the conditions, and all sit near the sampling-noise level. CB×SELF loses the fewest problems (23.0) and MEG the most (29.0), against 26.0 for VANILLA. Across the 27 runs, those that gained more distinct correct modes lost slightly more problems (Spearman $\rho = +0.40$, $p = 0.04$), but this does not survive correction over the 7 correlations we computed (Benjamini–Hochberg $p = 0.13$). Since the next subsection shows these modes are mostly wording, we treat it as a hypothesis.

D. What is lost: the human audit

H1, are the modes approaches? The annotators called 58 and 51 of the 60 pairs the same approach, although the partitioner had split half of them. They agreed on 51 pairs (85%), 50 SAME and 1 DIFFERENT. Cohen's κ is only 0.13 because almost every label falls in one class, which depresses κ however well raters agree; the prevalence-adjusted PABAK is $+0.70$. Against the consensus the partitioner scores 55%, below the 98% of a labeller that always says SAME. The pre-registered balanced accuracy is 77%, formally above the 70% bar, but one of its two classes holds a single pair, so we do not rely on it. We read the modes as surface clusters, and MEG as redistributing credit over them.

H2, what is lost? Vanilla GRPO lost 20 problems in a majority of its first three seeds (the audit was drawn before the extension runs). A and B disagreed on whether the base model's successes on them were valid: A rated 11 of 40 solutions valid and B 28 ($\kappa = -0.06$, agreement on 15 items). The third annotator labelled the 25 disputed items and sided with A on 22. By majority, 35% of exited solutions were valid (CI [18, 57], $n = 20$) against 25% of retained ones (CI [11, 47], $n = 20$; Fisher $p = 0.731$). Under the pre-registered rule this is *mixed*: most lost problems had been solved with flawed reasoning, but so had most kept ones, so vanilla GRPO did not selectively lose lucky hits or genuine solutions. (The 15 items A and B agreed on alone gave 5 of 6 against 2 of 9, $p = 0.041$.)

VI. Discussion and Limitations

Training intensity: a hypothesis. With about two rollouts per training problem and a policy that moved little (vanilla's Pass@1 rose by only 0.1 points; final KL to the base model 2–13 times its bf16 noise floor of about 2×10^{-4}), RLVR on a 0.5B model expands the boundary slightly. Accounts that tie shrinkage to repeated reinforcement of saturated problems [10] or to a long exploitation phase [9] predict exactly this, and the reported inversions [2], [3] come from far longer training. We therefore read our null as a statement about the regime a single-GPU budget reaches; we varied neither training length nor model size, so this remains a hypothesis. It suggests a direct test: raise rollouts per problem at 0.5B and shrinkage should appear. We registered that test (240-step runs scored at steps 120, 180 and 240) but could not afford it; our prediction, made before any such run, is that vanilla's slope declines over those steps, and if it does not, the intensity reading fails. **The partition is the weak link.** Embedding partitions put nearly every group in one mode (Table I), and the bigram partition finds about three modes that people see as one approach. Neither tracks what intra-group methods assume it tracks, which matters for any method that reweights credit by the diversity of correct rollouts within a group [14], [16], [17]. It is consistent with MEG ranking last on slope and net boundary gain: credit spread over wording variants does not buy new

approaches. A third of MATH training problems admit more than one approach [13], but a 0.5B model rarely samples them.

Limitations. One model at one size, LoRA, one RL algorithm and short training (120 steps, cut from a planned 240), evaluated only on MATH-500. Four or five seeds per cell leave the interactions open, and each run has one full evaluation, so we show no trajectory over training. SELF omits its forward-KL term and, by skipping solved prompts, takes fewer effective updates than vanilla. No random-partition control for MEG or random-weight control for CB was run. The annotators differed in strictness on H2, and the majority follows the stricter two. The correlations above are descriptive and were added after the pre-registration.

VII. Conclusion

In a pre-registered study of Qwen2.5-0.5B-Instruct with a blinded human audit, shrinkage does not appear at about two rollouts per training problem, and neither a topic-level budget, intra-group mode redistribution, nor their combination moves the shrinkage slope by more than about one point per unit $\ln k$. People judged the partitioner's modes to be wording variants of one approach. The protocol runs on one 24 GB GPU in 147 GPU-hours and is released.

Acknowledgment

We thank the three annotators. AWS credits came from the Amazon ML Challenge 2026. AWS compute: 147 A10G GPU-hours, about $148. An AI assistant (Claude, Anthropic) was used to write analysis code and to draft and edit parts of the text; the authors checked the results against the logs and take responsibility for the content.

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